

Day 5 Notes — Writing SOQL & DML in Apex

Salesforce Developer Bridge Program — practice notes with Developer Console query testing evidence

STUDENT	ROLL NO.	PROGRAM	DATE
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Mindset: Understand the business process first, then learn the code that implements it — "technology follows business."

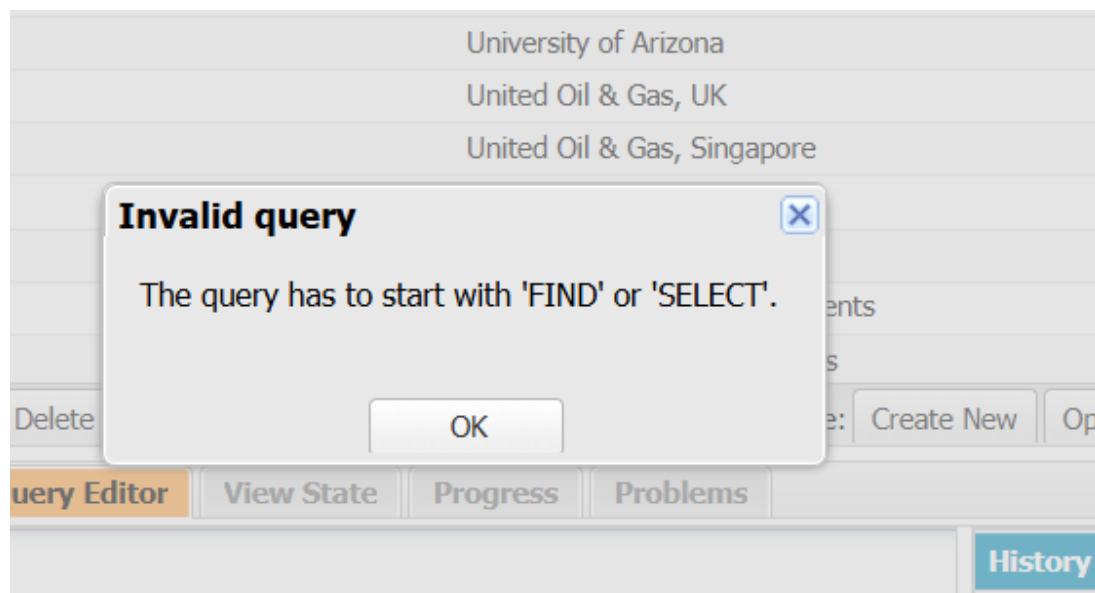
1. SOQL Syntax

General syntax:

```
SELECT field1, field2
FROM ObjectName
WHERE condition
LIMIT number
```

Keyword	Purpose
SELECT	Which fields do you need?
FROM	Which object contains the data?
WHERE	Which record(s) do you want?
LIMIT	Maximum number of records to return

Developer Console screenshot:



Practice: a query missing the `SELECT/FIND` keyword is rejected by the Query Editor — "The query has to start with 'FIND' or 'SELECT'." This confirms every SOQL query must begin with `SELECT` (or `FIND` for SOSL).

Example 1 — Get a Student

Business question: "What is this student's CGPA?"

```
Student__c student = [
SELECT Name, CGPA__c
FROM Student__c
WHERE Id = :studentId
LIMIT 1
];
```

Sample Output (illustrative, using sample data):

```
Student__c student = {
Name = 'Student 1',
CGPA__c = 8.70
}
```

Developer Console screenshot:

The screenshot shows the Salesforce Developer Console. At the top, there's a header with 'Account@9:58 AM', 'Student__c@10:14 AM', and 'Student__c@10:14 AM'. Below this, the query 'SELECT Id, Name, CGPA__c FROM Student__c LIMIT 5' is entered. The 'Query Results - Total Rows: 5' table is displayed with the following data:

Id	Name	CGPA__c
a00g500000p5G0RAAU	Kapil	
a00g500000p5mcwAAA	Student 1	
a00g500000p5mcwAAA	Student 2	
a00g500000p5mcyAAA	Student 3	
a00g500000p5mczAAA	Student 4	

Below the table, the 'Query Editor' tab is active, showing the same query: 'SELECT Id, Name, CGPA__c FROM Student__c LIMIT 5'. The 'History' panel on the right shows a list of executed queries, including the current one. At the bottom, there's an 'Execute' button and a checkbox for 'Use Tooling API'.

Practice: SELECT Id, Name, CGPA__c FROM Student__c LIMIT 5 — returned 5 rows with the CGPA__c field included, verifying this exact query pattern.

Example 2 — Get a Job

Business question: "What are this job's eligibility criteria?"

```
Job__c job = [
SELECT Name,
Minimum_CGPA__c,
Maximum_Backlogs__c
FROM Job__c
WHERE Id = :jobId
];
```

Sample Output (illustrative, using sample data):

```
Job__c job = {
Name = 'Software Engineer - Trainee',
Minimum_CGPA__c = 7.50,
Maximum_Backlogs__c = 0
}
```

Example 3 — Check Duplicate Application

Business question: "Has this student already applied?"

```
List<Application__c> apps = [  
  SELECT Id  
  FROM Application__c  
  WHERE Student__c = :studentId  
  AND Job__c = :jobId  
];
```

```
if(!apps.isEmpty()){  
  return 'Already Applied';  
}
```

Sample Output (illustrative, using sample data):

```
apps.size() = 0  
// isEmpty() = true -> condition skipped  
// Student 1 has not applied to this job yet
```

2. Bind Variables

The colon (:) before an Apex variable inside a SOQL query is called a **bind variable**. It tells Salesforce to use the value stored in that Apex variable.

```
WHERE Id = studentId // Invalid  
WHERE Id = :studentId // Correct
```

3. Query Results

- A query expected to return one record is assigned to a single sObject variable, e.g. Student__c student.
- A query expected to return many records is assigned to a list, e.g. List<Student__c> students.
- Use LIMIT 1 whenever only one record is expected.

```
SELECT Id, Name FROM Student__c LIMIT 10
```

Sample Output (illustrative, using sample data):

```
Total Rows: 10  
Kapil, Student 1, Student 2, Student 3, Student 4,  
Student 5, Student 6, Student 7, Student 8, Student 9
```

Developer Console screenshot:

Account@9:58 AM | Student__c@10:14 AM

SELECT Id, Name FROM Student__c LIMIT 10

Query Results - Total Rows: 10	
Id	Name
a00g500000p5G0RAAU	Kapil
a00g500000p5mcwAAA	Student 1
a00g500000p5mcwAAA	Student 2
a00g500000p5mcwAAA	Student 3
a00g500000p5mczAAA	Student 4
a00g500000p5md0AAA	Student 5
a00g500000p5md1AAA	Student 6
a00g500000p5md2AAA	Student 7
a00g500000p5md3AAA	Student 8
a00g500000p5md4AAA	Student 9

Query Grid: Save Rows Insert Row Delete Row Refresh Grid Access in Salesforce: Create New Open Detail Page Edit Page

Logs Tests Checkpoints **Query Editor** View State Progress Problems

```
SELECT Id, Name
FROM Student__c
LIMIT 10
```

Any query errors will appear here...

History

- Executed
- SELECT id, Name FROM Account
- SELECT Id, Name FROM Student__c

Practice: `SELECT Id, Name FROM Student__c LIMIT 10` — returned 10 rows (Kapil and Student 1-9), confirming the list-of-records pattern.

Developer Console screenshot:

SELECT Id, Name FROM Student__c LIMIT 5

Query Results - Total Rows: 5	
Id	Name
a00g500000p5G0RAAU	Kapil
a00g500000p5mcwAAA	Student 1
a00g500000p5mcwAAA	Student 2
a00g500000p5mcwAAA	Student 3
a00g500000p5mczAAA	Student 4

Query Grid: Save Rows Insert Row Delete Row Refresh Grid Access in Salesforce: Create New Open Detail Page Edit Page

Logs Tests Checkpoints **Query Editor** View State Progress Problems

```
SELECT Id,
Name
FROM Student__c
LIMIT 5
```

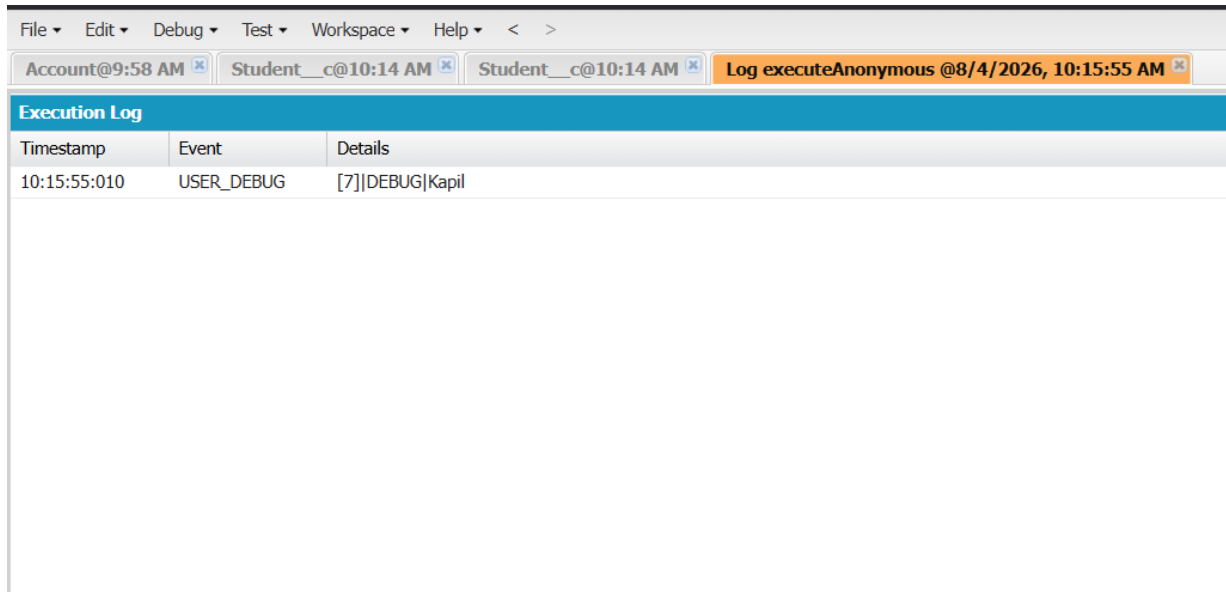
Any query errors will appear here...

History

- Executed
- SELECT id, Name FROM Account
- SELECT Id, Name FROM Student__c
- SELECT Id, Name, CGPA__c FROM...
- SELECT Id, Name, CGPA__c FROM...
- SELECT Id, Name FROM Student__c

Practice: the same `LIMIT 5` query written across multiple lines — Salesforce SOQL ignores line breaks/whitespace, so the result set is identical.

Developer Console screenshot:



Practice: running `executeAnonymous` with a debug statement on the retrieved record — the Execution Log's `USER_DEBUG` line prints 'Kapil', confirming the `Name` field was read correctly.

4. DML Statements

Insert — creates a record

```
Student__c student = new Student__c();
student.Name='Datta';
insert student;
```

Sample Output (illustrative, using sample data):

```
New record created:
Id = a00g500000p5newAAA
Name = 'Datta'
```

Flow: Create Object → Assign Values → Insert

Update — retrieve, modify, then update

```
student.CGPA__c = 8.5;
update student;
```

Sample Output (illustrative, using sample data):

```
Record a00g500000p5newAAA updated:
CGPA__c = 8.5
```

Delete — removes the record

```
delete student;
```

Sample Output (illustrative, using sample data):

```
Record a00g500000p5newAAA deleted
```

5. Creating a Record

```
Application__c app = new Application__c();
app.Student__c = studentId;
app.Job__c = jobId;
app.Status__c = 'Applied';
insert app;
```

Or, using a constructor:

```
Application__c app = new Application__c(
    Student__c = studentId,
    Job__c = jobId,
    Status__c = 'Applied'
);
insert app;
```

Both approaches are valid.

6. Typical Enterprise Flow

Every service generally follows this structure. SOQL and DML are not isolated steps — they are part of one business transaction.

```
Receive Request
↓
Retrieve Records (SOQL)
↓
Validate Rules
↓
Create/Update Record
↓
Save (DML)
↓
Return Result
```

7. Building submitApplication()

Step 1 — Receive IDs

```
public static String submitApplication(Id studentId, Id jobId)
```

Step 2 — Retrieve Student

```
Student__c student = [
    SELECT CGPA__c, Branch__c
    FROM Student__c
    WHERE Id = :studentId
];
```

Step 3 — Retrieve Job

```
Job__c job = [  
SELECT Minimum_CGPA__c, Maximum_Backlogs__c  
FROM Job__c  
WHERE Id = :jobId  
];
```

Step 4 — Check Duplicate

```
List<Application__c> apps = [  
SELECT Id  
FROM Application__c  
WHERE Student__c = :studentId  
AND Job__c = :jobId  
];  
  
if(!apps.isEmpty()){  
return 'Duplicate';  
}
```

Step 5 — Validate

```
if(student.CGPA__c < job.Minimum_CGPA__c){  
return 'CGPA Not Eligible';  
}  
// Similarly validate: Backlogs, Branch, Deadline
```

Step 6 — Create Record

```
Application__c app = new Application__c();  
app.Student__c = studentId;  
app.Job__c = jobId;  
app.Status__c = 'Applied';
```

Step 7 — Insert

```
insert app;
```

Step 8 — Return

```
return 'Application Submitted Successfully';
```

Complete Flow

```
Student Clicks Apply
↓
submitApplication(studentId, jobId)
↓
SOQL → Student
↓
SOQL → Job
↓
SOQL → Existing Application
↓
Business Validation
↓
Create Application Object
↓
insert Application
↓
Success Message
```

8. Common Interview Questions

Q: What is a bind variable?

A bind variable is an Apex variable referenced inside a SOQL query using a colon, e.g. WHERE Id = :studentId.

Q: Why use LIMIT 1?

When only one record is expected, LIMIT 1 restricts the query to a single result.

Q: Difference between SOQL and DML?

SOQL	DML
Retrieves data	Modifies data
SELECT	insert, update, delete, undelete
Used before validation	Used after validation

Q: Why validate before insert?

Saving invalid data can lead to incorrect business records, duplicate applications, and broken business rules — validation is always placed before any DML operation.

9. Day-5 Final Mindset

Don't memorize SOQL or DML syntax in isolation. Think in terms of the business workflow:

1. Ask a business question → SOQL retrieves the required information.
2. Apply business rules → Apex decides what should happen.
3. Change business data → DML inserts or updates records.
4. Communicate the outcome → Return a meaningful result to the user.